WSC 2026 talk: reading pack

15 October 2026 · Sebastian Fridman · offline copy of the Claude Doc "WSC 2026 talk: reading pack"

Contents: Start here · Reproducibility · NINDS rt-PA · Other re-analyses · Readers and citations · Imaging tools · Value · Proposals · Questions

Start here

Eight sources, about two hours in total, cover what you are most likely to be challenged on: the NINDS story, the citation claim, open imaging tools and the cost of "on request". Every number on the new slides traces to one of them or to the sections below.

#	Source	Why read it	Slide	Time
1	Dachs 2008, PLoS Med	How the NINDS data were refused, then sold on CD in 2003; the 27 reports by trial authors meanwhile	NINDS rt-PA	10 min
2	Ingall 2004, Stroke	The independent re-analysis: OR 2.1 after adjustment	NINDS rt-PA	15 min
3	UK MHRA expert group 2015, pages 9 to 12	How a regulator weighed the critics' and the committee's methods	NINDS rt-PA	15 min
4	Lindley, Wardlaw, Sandercock 2005, Lancet Neurol	Why the blood-pressure recommendations from NINDS needed reconsidering	NINDS rt-PA	10 min
5	Davis 2008, BMJ	The randomised trial of open access: more downloads, no more citations	Access	10 min
6	de la Rosa 2025, Nat Commun	DeepISLES: three open challenge models in one tool, validated on 1,685 scans	Access	20 min
7	Scutt 2020, BMJ Open	Bath's group asked 391 vascular and cognitive trials for data: 20% delivered, median 241 days	Value, data statements	10 min
8	Taherifard 2026, BMJ	336 shared trials: 79% reused, 209 papers by outside teams	Value	10 min

Reproducibility: the crisis next door

The four historical numbers measure different things (replication, power, publication bias in animal stroke studies, experience), so present them side by side, never as one failure rate. Ten years on, psychology has rules but practice still lags.

Source	What it shows	On the slide	Watch out
Open Science Collaboration 2015, Science	36% of 100 psychology findings replicated with a significant result; effects about half the size	36%	Replication in new data, not reanalysis
Button 2013, Nat Rev Neurosci	Median statistical power of neuroscience studies 21%	21%	A power estimate, not a failure rate
Sena 2010, PLoS Biol	Of 525 animal stroke publications, only 10 (2%) reported no effect on infarct volume; publication bias explains about a third of reported efficacy (31.3% to 23.8%); about 14% of experiments never published	10 of 525	Publication bias, not replication
O'Collins 2006, Ann Neurol	1,026 experimental stroke treatments: the 114 used clinically were no more effective in animals than the 912 tested only in animals	Notes	"None translated" is not in the abstract; do not say it
Macleod 2008, Stroke	NXY-059 improved infarct volume by 43% in animals, but only 2 of 9 publications reported randomisation, concealment and blinding; SAINT I positive, SAINT II neutral	Notes	
Baker 2016, Nature	70% of 1,576 researchers had failed to reproduce another group's experiment	70%	A survey of experience
SCORE, Nature 2026 (talk/evidence_intro.md)	55% of 274 social and behavioural science claims replicated; effect r 0.25 to 0.10; data available for 24% of 600 papers	55%	Social and behavioural sciences, not medicine
Hardwicke 2024, AMPPS	7% of 2022 psychology articles preregistered, 14% shared raw data; 3% and 2% in 2014 to 2017	7% and 14%	
Begley & Ellis 2012, Nature	Amgen confirmed 6 of 53 landmark preclinical cancer findings	Off the slide; if asked	A company report; the data were never shared
Simmons 2011, Psychol Sci	Four routine analysis choices raise a 5% false-positive rate to 61%	Notes	Simulation
John 2012, Psychol Sci	63% of 2,155 psychologists admitted not reporting all their measures	Notes	Self-report
National Academies 2019	Definitions: reproducibility (same data and code) vs replicability (new data)	Notes	Use these terms if challenged

If asked about lesion mapping: van den Heuvel 2026 (Nat Neurosci, doi) re-analysed lesion network mapping with public data only; replies include Zalesky and Cash, Siddiqi and Fox, and Corbetta's group. Present it as a live debate that open data made possible.

NINDS rt-PA: the data story

Opening the NINDS data did not sink tPA: an independent re-analysis confirmed the benefit (OR 2.1) and found what the papers had not said. The table runs oldest first because it is a story.

Year	What happened	Source
1995	Two-part trial, 624 patients. Part 1 missed its 24-hour primary outcome; part 2 global OR 1.7 (1.2 to 2.6). Symptomatic haemorrhage 6.4% vs 0.6%; mortality 17% vs 21% (not significant)	NEJM 1995
1995 to 2003	Critics cite an imbalance: NIHSS 0 to 5 in 42 tPA vs 16 placebo patients. Outside researchers are refused the data, even after a Freedom of Information petition to the FDA; trial authors publish at least 27 reports from the same dataset	Dachs 2008; MHRA 2015
2002	BMJ investigation (Lenzer, March); in November: "Controversial stroke trial is under review following BMJ report". NINDS commissions an independent committee	Lenzer 2002
2003	Data released: NTIS CD-ROM, US$79 plus shipping, 100 variables, SAS conversions, no codebook. By 2008 only three outside groups had used it	Dachs 2008
2004	Committee: benefit holds after adjustment, OR 2.1 (1.5 to 2.9). New: no time gradient within 3 h detectable; blood-pressure data "too unreliable for inclusion in analysis" (112 patients, 18%, with identical admission and randomisation readings; 22 missing). Committee members had no connection with NINDS, the trial or the manufacturer; it still recommended rigorous BP management	Ingall 2004; Hertzberg 2008
2005	Imbalance did not materially affect the result; BP recommendations from NINDS need reconsidering	Lindley 2005
2007	Severity-adjusted endpoint confirms benefit (public data)	Saver & Yafeh 2007
2009	Graphs of all 624 patients: differences "diminish or disappear" on change in NIHSS; calls for larger trials	Hoffman & Schriger 2009
2010	Matched re-analysis: benefit persists, 13 to 23% smaller, mostly in large-vessel occlusion. Graphic re-analysis: tPA patients recover two-thirds of the possible change vs half with placebo	Mandava 2010; Saver 2010
2015	UK regulator's expert group: change-from-baseline is biased against the arm with more mild strokes; randomisation errors had no impact (FDA analysis); half the 0 to 90 min patients were treated at 89 to 90 min, so NINDS cannot test time-is-brain; no regulatory implications	MHRA 2015
2022 to 2023	Selection-bias claim (3 of 7 positive results lose significance; all ORs still favour tPA). Eight investigators reply: every error disclosed to FDA, most gave placebo to patients assigned tPA; "All data from the original trial is available from the NINDS"	Garg 2022; Lyden 2023
2026	NINDS archive: "data requests will no longer be accepted … contact the Principal Investigators"; migration expected by end of 2026	NINDS page, checked 15 Oct 2026

Two cautions for the room: trial investigators often attend WSC, and the critics' conclusions are not settled science, so describe them as "critics re-analysed the same file". The BMJ 2002 claims about Genentech funding were not verified and stay off the slides.

Other re-analyses by outsiders

Independent re-analyses are rare because access is rare: of 37 published re-analyses of trial data, only 5 were by fully independent teams, and where data were shared, 14 of 17 trials reproduced. The argument for openness is verification, with occasional correction.

Case	What was open	What outsiders found	Caveat
Cluster failure, Eklund 2016, PNAS	499 resting-state scans from the 1000 Functional Connectomes Project	fMRI cluster inference gave false positives up to 70% against a nominal 5%; a 15-year-old AFNI bug	70% is the worst case; the "40,000 studies" sentence was corrected
ECASS III, Alper 2020, BMJ EBM	Boehringer data through ClinicalStudyDataRequest.com	Published significance held only under three choices not in the protocol; seven alternatives gave P 0.054 to 0.173	Contested; guidelines unchanged; authors had argued for re-analysis before. Backup slide at most
Rosiglitazone, Nissen 2007, NEJM and Wallach 2020, BMJ	Results GSK had to post after a 2004 settlement; later patient data via CSDR	MI OR 1.43, then FDA and EU restrictions (2010); patient data showed more MIs than published summaries in 26 of 29 trials	FDA lifted restrictions in 2013; size of the risk still debated
Study 329, Le Noury 2015, BMJ	GSK documents and patient data, external team	No efficacy; suicidal or self-injurious behaviour 11/4/2 vs 5/3/1 published	Original paper not retracted; expression of concern 2026
Tamiflu, Jefferson 2014, BMJ	Clinical study reports from EMA and Roche, about 150,000 pages	Symptoms 16.8 h shorter; no effect on hospital admission	A Roche-funded analysis disagrees (Dobson 2015)
Sato trials, Bolland 2016, Neurology	Published baseline tables only	Fabrication in trials of hemiplegic stroke patients; about 30 retractions incl. JAMA, Neurology, Stroke	Forensics of summaries, not shared data
Surgisphere 2020: Lancet retraction, NEJM retraction	Nothing	Two COVID papers retracted within weeks because no auditor could see the data; the Lancet then required data-sharing statements	Fabrication never formally adjudicated
Ebrahim 2014, JAMA	37 published re-analyses	13 (35%) changed who or how many to treat; only 5 by independent authors	Mostly the original authors re-analysing
Naudet 2018, BMJ	37 RCTs in journals with full sharing policies	Data obtained for 17 (46%); 14 of 17 fully reproduced	Sharing policies are not sharing

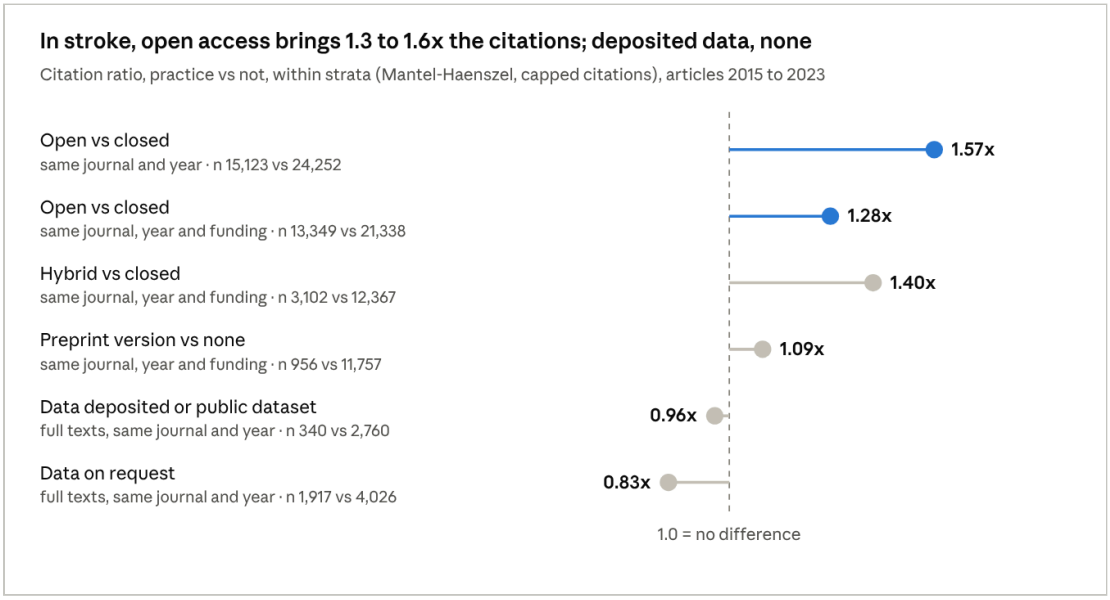
One more if you want a cardiology example of shared data correcting a paper: in the SPRINT data challenge, a Yale team found the published cardiovascular risk profile was wrong (76%, not 61%, at 15% or more risk) and NEJM corrected it (Warner 2018).

Access: readers and citations

Open work is read more and cited somewhat more; sharing data earns no citations. Say "readers", not "citations": the strongest designs found more downloads and no extra citations.

Source	Design	Finding	Watch out
Davis 2008, BMJ	Randomised: 1,619 articles, 11 journals	Full-text downloads +89% (76 to 103%); 59% vs 63% cited at 1 year	Physiology journals
Davis 2011, FASEB J	Randomised: 36 journals	Cited "no more frequently, nor earlier" at 3 years	
Langham-Putrow 2021, PLoS One	Systematic review, 134 studies	47.8% found an open-access advantage, 27.6% none, 23.9% only in subsets	About 98% at high risk of bias
Piwowar 2018, PeerJ	100,000 articles	Open 1.18 vs world average, closed 0.90, gold 0.83	"+18%" is vs world average, not vs closed
Piwowar & Vision 2013, PeerJ	10,555 microarray studies	Shared data +9% (5 to 13%) after many controls	Genomics
Colavizza 2024, PLoS One	About 122,000 papers	Data in a repository +4.3%; preprints +20%; code no advantage	Mostly PLOS, where sharing is mandatory
Fu & Hughey 2019, eLife	74,239 bioRxiv-era articles	Preprinted articles 1.36x citations	Biology
Zissette 2024, JAMA Netw Open	5,739 articles in 25 clinical journals	Preprints: no difference outside COVID	

Our stroke data (first stroke-specific estimate found). 2015 to 2023 articles, OpenAlex citations, compared within the same journal and year; funded papers are 41% of open and 20% of closed articles. Table: `talk/outputs/tables/stroke_citations_by_open_practice.csv`; code: `citation_outputs` in `scripts/wsc2026_talk_analysis.py`.



Matching on funding halves the open-access advantage; a deposited dataset or an "on request" statement brings no extra citations. Observational: authors may open their stronger work.

Access: imaging tools

Commercial stroke AI works where hospitals can pay for it; open algorithms exist, but none is regulated or tested in a low- or middle-income country. Frame this as a business-model problem, not a company problem: JLK and Heuron are Korean vendors.

Theme	Source	Finding
Benefit	Martinez-Gutierrez 2023, JAMA Neurol	Viz.ai stepped-wedge cluster RCT: door-to-groin -11.2 min (-18.2 to -4.2); no mRS difference. PI received Viz grants
Benefit	Nagaratnam 2025, Lancet Digit Health	Brainomix at 26 English hospitals: thrombectomy 2.3% to 4.6% vs 1.6% to 2.6% elsewhere. Observational, company co-authors
Cost	NICE HTG708, 2024	Licences about GBP 15,000 to 30,000 a year per centre (manufacturers' estimates)
Cost	US CMS final rule FY2021 (85 FR 58636)	Add-on payment for Viz.ai ContaCT up to $1,040 per case
Cost, Korea	HIRA AI guideline 2023; press (Hitnews, Sep 2024)	Stroke AI not covered by national insurance; non-contrast CT LVO screening capped at KRW 18,100 per scan (press only)
Dependence	Albers 2018, NEJM (DEFUSE 3)	Core and penumbra "calculated with the use of RAPID software"; DAWN used the same. FDA K182130 (2018) then cleared RAPID for thrombectomy selection
Disagreement	Mallon 2022, J Stroke Cerebrovasc Dis	Brainomix vs RapidAI, same 54 patients: thrombectomy eligibility discordant in 15%
Disagreement	Koopman 2019, JNIS	Same scans: median core 50 mL (Philips), 41 mL (Siemens), 11 mL (RAPID)
Agreement	Pisani 2023, Ann Neurol	Newer packages mostly agree: RAPID vs Viz core rho 0.77
Real world	Kunst 2024, JACR	LVO triage sensitivity 97% (manufacturer) fell to 85% for ICA/M1, 69% with M2
Simpler imaging	MR CLEAN-LATE 2023, Lancet; CLEAR 2022, JAMA Neurol	Late-window selection by CTA collaterals works; NCCT vs CTP selection, no difference in 90-day mRS
Access gap	MT-GLASS, Asif 2023, Circulation	Median thrombectomy access 2.79% of LVO patients; none in 7 countries; 88% lower in low-income countries
Access gap	Hricak 2021, Lancet Oncol	One CT scanner per 25,000 people (high income) vs 1.7 million (low income)
Access gap, Korea	Lee 2026, BMC Health Serv Res	22.5% of ischaemic stroke patients (2021) lived in a district without a thrombectomy centre

Theme	Source	Finding
Open	de la Rosa 2025, Nat Commun	DeepISLES, Apache-2.0, 1,685 external scans, preferred to manual masks. MRI only, research use
Open	Brugnara 2023, Nat Commun	Academic LVO detector on an open framework beat two cleared products by 25 to 45% in sensitivity. Partly open
Gap	ISLES'24 (preprint, arXiv 2408.10966)	Final infarct from acute CT and perfusion: best Dice 0.285. No open CTP or LVO tool has peer-reviewed validation
Limits	EU MDR Article 5(5)	An in-house device cannot be passed to another hospital without a legal manufacturer

Value

Shared data multiply what public money buys; closed data cost time and follow-on research. The waste estimate is the softest number here, so call it an estimate.

Source	Finding	Slide	Watch out
Chalmers & Glasziou 2009, Lancet; their 2016 BMJ blog	85% of health research investment estimated wasted: about half lost at each of design, publication and reporting	85%	An estimate built from stage losses; verified only through the authors' blog
Macleod 2014, Lancet	Opened the series "Research: increasing value, reducing waste"; authors include Macleod and Al-Shahi Salman (Edinburgh) and Dirnagl (Berlin)	Notes	A stroke hook, not a number
Taherifard 2026, BMJ	336 J&J trials shared via YODA: 265 (79%) reused, 1,167 papers, 209 by outside teams; outside share passes half by year 11	209	Outside papers were cited less and less often in guidelines
Milham 2018, Nat Commun	Shared neuroimaging data used in 913 papers by authors in 50 countries; collecting them again would cost $893 million to $1.71 billion	$0.9 to 1.7B	Cost per subject assumed, $1,000 to $10,000
Williams 2013, J Polit Econ (NBER version)	Genes held under Celera's licence for up to two years: 20 to 30% less follow-on research and product development	20 to 30%	Genomics, not clinical trials
Scutt 2020, BMJ Open	Of 391 vascular and cognitive trials asked for data, 78 (20%) delivered; median 241 days	Notes, data statements	The closest "on request" evidence to stroke

Background for the proposals

Rules alone produced statements, not data; incentives and infrastructure are what moved practice elsewhere. Most toolkit pieces already exist, so the ask to WSO is stewardship, not invention.

Proposal	Source	Finding
Task force	Gabelica 2022, J Clin Epidemiol	Of 1,792 authors promising data "on request", 123 (6.8%) shared
Task force	Danchev 2021, JAMA Netw Open	Of 334 JAMA, Lancet and NEJM RCTs declaring sharing, 2 (0.5%) had downloadable data
Task force	Bergeat 2022, JAMA Netw Open	ICMJE statement rule in surgical trials: data obtainable for 3.1% before and 3.1% after
Task force	Scheel 2021, AMPPS; Soderberg 2021, Nat Hum Behav	Registered Reports: 44% positive results vs 96%; rated higher on rigour, equal on novelty
Task force	ICMJE (NEJM 2017)	Data-sharing statements required since July 2018; the 2016 draft mandating patient-data release within 6 months was dropped
Task force	Evidence brief (talk/evidence_brief.md)	No WSO or ESO position statement on open science was found; no stroke journal offers Registered Reports
Toolkits	IST-3 data, Edinburgh DataShare	Individual patient data open since January 2021, no application needed: the stroke model to cite
Toolkits	VISTA	Its rules exclude re-analysis of treatment effects and promise contributors "no risk of re-analysis": not a model for open trial data
Toolkits	MR CLEAN Registry (evidence brief)	Patient data cannot be shared under Dutch law because consent for sharing was not obtained: consent wording belongs in the kit
Toolkits	NINDS stroke common data elements, Saver 2012; utility-weighted mRS, 2015	Free standard data elements; published utility weights (1.0 / 0.91 / 0.76 / 0.65 / 0.33 / 0 / 0)
Toolkits	DeepISLES, nnU-Net, BIDS	Open imaging pieces that exist today (see Access: imaging tools)
Prize	Kidwell 2016, PLoS Biol; Rowhani-Farid 2020	Badges: Psychological Science open data from under 3% to 39%; the only medical RCT, OR 0.9
Prize	Rice 2020, BMJ	0 of 92 biomedical promotion guidelines mention data sharing, registration or open access
Prize	Einstein Foundation Award (evidence brief)	EUR 500,000 a year since 2021 for promoting quality in research: the model for a WSO prize
Prize	Cobey 2024, Open Heart	Cardiology researchers (n = 198) named funding as the most essential incentive

Questions to expect

Most hard questions are about method limits or about tPA; answer the limit first, then the number.

Question	Short answer	Where
Did the NINDS re-analysis show tPA does not work?	No. The independent committee found the benefit held after adjustment (OR 2.1), and the UK regulator saw no regulatory implications. Critics re-analysed the same file; that debate was possible only because the data were released	NINDS section
Is your corpus from PubMed?	No: an OpenAlex title search, journal articles and reviews with a DOI. A wider title-and-abstract search for 2024 gave 37,878 vs 15,602 articles and the same open-access rate, 75.5% vs 75.2%	talk/outputs/tables/corpus_sensitivity_2024.csv
Abstracts undercount data and code sharing	Yes, so abstract rates are lower bounds. That is why we read 16,362 full texts: GitHub appears in 1.3% of full texts vs 0.07% of abstracts	Method, Data and code slides
The full-text audit only covers open-access articles	Yes, so its rates are optimistic for stroke overall; up to 1,500 per year, each year weighted the same	Method slide notes
Will sharing get me more citations?	Open access: about 1.6x in the same journal and year, 1.3x at equal funding. Deposited data: no difference. The randomised trial found more readers, not more citations	Access: readers
Privacy and consent stop us sharing	Controlled-access platforms exist (Vivli, YODA); consent wording is the fix, as MR CLEAN Registry shows; IST-3 shares patient data openly	Background for the proposals
Commercial imaging AI is regulated and works	Agreed: a cluster RCT and English data show benefit. The gap is cost and access, and that no open tool is yet regulated or validated where most strokes happen	Access: imaging tools
Is the commitment index a quality ranking?	No. A descriptive mean of deposit, GitHub and registration rates in the full text, relative to the stroke average; small counts move the order	Commitment slide notes
Why is neuroscience ahead on code and preprints?	Partly its title term "neural", about 45% of that corpus, also catches machine-learning papers	Method slide notes
Who pays for the toolkits and the prize?	The position statement, retiring "on request" and promotion criteria cost nothing. Registered Reports, the prize and the toolkits need money; most tools already exist	Proposals slide
What about industry trials?	Platforms work: 79% of 336 J&J trials shared via YODA were reused, 209 papers by outside teams	Value